

231501075

EX:2

EDA-Data Import and Export

AD23632

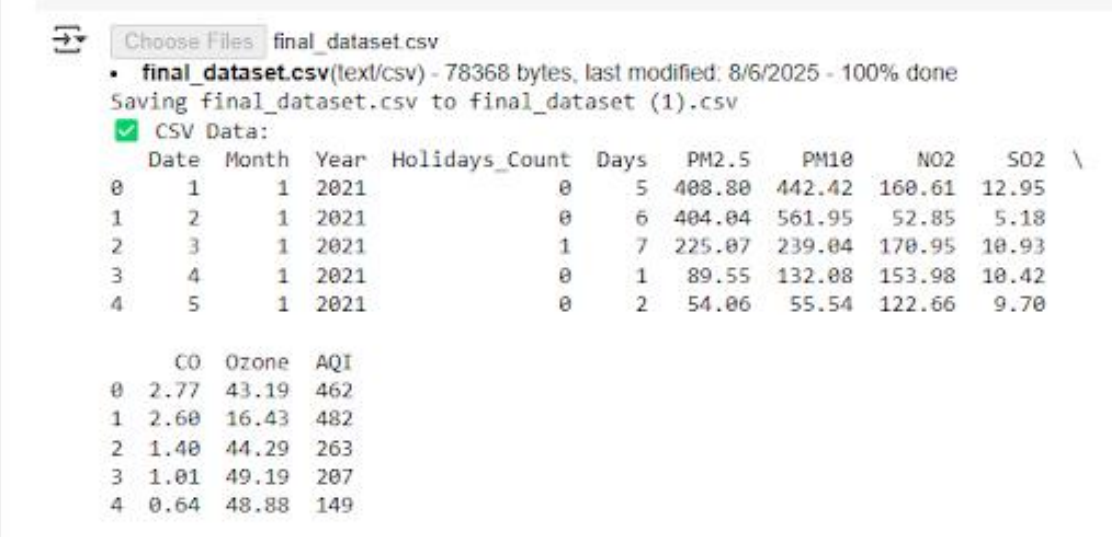
Aim:

- Importing data from CSV, Excel, SQL databases, and web scraping
- Handling different data formats
- Export a DataFrame to an Excel file

CODE:

```
import pandas as pd
import sqlite3
from google.colab import files

uploaded = files.upload()
df_csv = pd.read_csv('final_dataset.csv')
print("✓ CSV Data:")
print(df_csv.head())
```



The screenshot shows a Google Colab interface. At the top, there is a 'Choose Files' button and a file named 'final_dataset.csv' is listed. Below this, a message states: 'final_dataset.csv(text/csv) - 78368 bytes, last modified: 8/6/2025 - 100% done'. A green checkmark indicates the file is saved as 'final_dataset (1).csv'. The main part of the screenshot displays the 'CSV Data:' as a table with two sections. The first section has columns: Date, Month, Year, Holidays_Count, Days, PM2.5, PM10, NO2, and SO2. The second section has columns: CO, Ozone, and AQI. The data rows are indexed from 0 to 4.

	Date	Month	Year	Holidays_Count	Days	PM2.5	PM10	NO2	SO2	
0	1	1	2021	0	5	408.80	442.42	160.61	12.95	
1	2	1	2021	0	6	404.04	561.95	52.85	5.18	
2	3	1	2021	1	7	225.07	239.04	170.95	10.93	
3	4	1	2021	0	1	89.55	132.08	153.98	10.42	
4	5	1	2021	0	2	54.06	55.54	122.66	9.70	

	CO	Ozone	AQI
0	2.77	43.19	462
1	2.60	16.43	482
2	1.40	44.29	263
3	1.01	49.19	207
4	0.64	48.88	149

```
uploaded = files.upload()
df_excel = pd.read_excel('converted_file.xlsx')
print("✓ Excel Data:")
print(df_excel.head())
```

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Choose Files converted_file.xlsx

• converted_file.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 1190399 bytes, last modified: 8/6/2025 - 100% done

Saving converted_file.xlsx to converted_file (2).xlsx

✓ Excel Data:

	letter	xbox	ybox	width	height	onpix	xbar	ybar	x2bar	y2bar	\
0	I	2	8	3	5	1	8	13	8	6	
1	I	5	12	3	7	2	10	5	5	4	
2	0	4	11	6	8	6	10	6	2	6	
3	N	7	11	6	6	3	5	9	4	6	
4	G	2	1	3	1	1	8	6	6	6	

	xybar	x2ybar	xy2bar	xedge	xedgey	yedge	yedgey
0	6	10	8	0	8	0	8
1	13	3	9	2	8	4	10
2	10	3	7	3	7	3	9
3	4	4	10	6	10	2	8
4	6	5	9	1	7	5	10

```
conn = sqlite3.connect(':memory:')
cursor = conn.cursor()
```

```
cursor.execute("CREATE TABLE products (id INTEGER, name TEXT, price
INTEGER)")
cursor.executemany("INSERT INTO products VALUES (?, ?, ?)", [
    (1, 'Laptop', 1000),
    (2, 'Tablet', 600),
    (3, 'Phone', 400)
])
```

```
conn.commit()
```

```
df_sql = pd.read_sql_query("SELECT * FROM products", conn)
print("✓ SQL Data:")
print(df_sql)
```

✓ SQL Data:

	id	name	price
0	1	Laptop	1000
1	2	Tablet	600
2	3	Phone	400

```
df_web =
pd.read_html("https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nomi
nal)")[2]
print("✓ Web-Scraped Table:")
print(df_web.head())
```

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Web-Scraped Table:

Country/Territory IMF[1][12]		World Bank[13]	
Country/Territory	Forecast	Year	Estimate
0 World	113795678	2025	111326370
1 United States	30507217	2025	29184890
2 China	19231705	[n 1]2025	18743803 [n 3]2024
3 Germany	4744804	2025	4659929
4 India	4187017	2025	3912686

United Nations[14]	
Estimate	Year
0 100834796	2022
1 27720700	2023
2 17794782	[n 1]2023
3 4525704	2023
4 3575778	2023

```
json_data = '{"name": ["Alice", "Bob"], "age": [25, 30]}'
df_json = pd.read_json(json_data)
print("✓ JSON Data:")
print(df_json)
```

✓ JSON Data:

	name	age
0	Alice	25
1	Bob	30

/tmp/ipython-input-2620177547.py:2: FutureWarning: Passing literal json to 'read_json'
df_json = pd.read_json(json_data)

```
data_dict = {
    'Fruit': ['Apple', 'Banana', 'Orange'],
    'Quantity': [10, 15, 20]
}
df_dict = pd.DataFrame(data_dict)
print("✓ Dictionary Data:")
print(df_dict)
```

✓ Dictionary Data:

	Fruit	Quantity
0	Apple	10
1	Banana	15
2	Orange	20

```
df_dict.to_excel("exported_file.xlsx", index=False)
print("✓ DataFrame exported to Excel: exported_file.xlsx")
```

✓ DataFrame exported to Excel: exported_file.xlsx

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Result:

Thus the EDA-Data Import and Export is done successfully.